Final PROJECT TITLE

Hand Gesture-Based Communication System for Disabled Individual

Panel Head

	Dr. R.Babu

Project Domain

Passport size photo(s)

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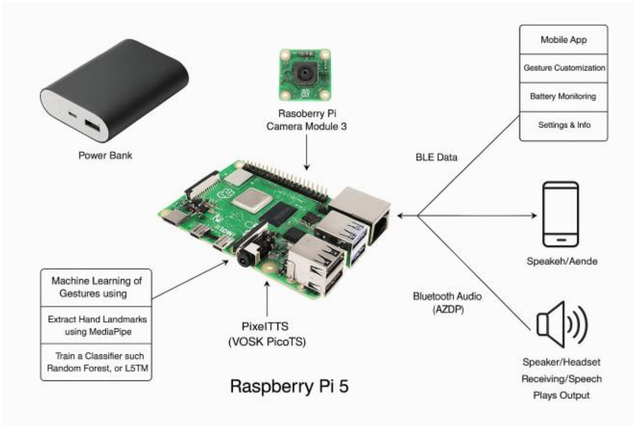
Abstract

Hearing-impaired communication with the rest of society continues to be a chronic stumbling block in society, usually leading to exclusion, misperception, and compromised access to critical services. Conventional mechanisms like interpreters or written forms are not always readily available or feasible in everyday conversations. To minimize this gap, this project suggests the development and deployment of a real-time sign language interpreter system based on low-cost, portable, and edge-based hardware — namely the Raspberry Pi 5.

Architecture Diagram



Significance of the Project

GestureTalk integrates gesture recognition, real-time translation, speech output, and accessibility support within one seamless solution designed for simplicity. Unlike other systems that require multiple tools or devices to interpret sign language, our project offers a unified communication platform

Conclusion

GestureTalk addresses social inclusion by empowering disabled individuals with a reliable communication tool. It bridges the communication gap, fosters independence, and enhances confidence. The system contributes to inclusive education, accessible healthcare, and improved quality of life, especially in underserved areas where sign language interpreters are unavailable. It also promotes awareness and empathy towards the disabled community.

Conference/Journal Publication Details (Mandatory)

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